Building a Daily DTD Web Service

Background

You are joining a small team responsible for maintaining a **real-world, production credit risk system**.

The system **runs daily in production**, with a recurring workflow that includes:

1. ingesting new market, risk-free rate and financial statement data,
2. preparing model inputs and computing AV and DTD using supplied parameters,
3. validating results for production use,
4. and delivering outputs to users through a simple web service.

One of the core outputs of this system is **Distance-to-Default (DTD)**, derived from the Merton structural credit risk framework. DTD is treated as a **signal produced by a system**, not as a standalone analytical result.

This task is designed to assess whether you can:

- understand the **financial meaning** of a credit risk signal,
- translate that understanding into **data pipelines and validation logic**, and
- reason about problems from a **production system perspective**, rather than as one-off analyses.

Task Structure Overview

Build an automated daily DTD service, with an understanding and reflection component. The task is divided into two parts:

1. **Daily DTD Service:** Integrate data preparation, DTD computation and a simple web interface into one working system. Support daily ingestion and batch processing through the same pipeline.
2. **Understanding and Reflection:** Interpret DTD, compare results and investigate potential issues.

Data Provided by CRI

Firm context. The sample company is Greentown China Holdings Limited (03900.HK), company ID 23286. Historical inputs cover **2 January 2024 to 31 December 2025** and come from CRI's daily production pipeline.

Historical Inputs

Daily inputs for 2024–2025, to be extended with newly collected data.

Input column	Description
`Comp_no`	Company identifier
`Date`	Trading date, YYYYMMDD
`CUR_MKT_CAP(HKD)`	Daily market capitalization, HKD million
`BS_CUR_LIAB(HKD)`	Current liabilities, HKD million
`BS_LT_BORROW(HKD)`	Long-term borrowings, including non-current lease liabilities, HKD million
`BS_TOT_LIAB2(HKD)`	Total liabilities, HKD million
`BS_TOT_ASSET(HKD)`	Total assets, HKD million; retained for reference, not required for computation
`Risk_Free_Rate`	Annual risk-free rate in percentage units (2.97 means 2.97%)

Parameters

Dated asset volatility σ and other-liability weight δ are provided through the latest available trading date; no parameter estimation is required.

Parameter column	Description
`COMPANY_ID`	Company identifier
`REAL_DTD_DATE`	Date of the parameter record, YYYYMMDD
`DTD_SIGMA`	Annualized asset volatility σ, in decimal units
`DTD_WEIGHT`	Other-liability weight δ

Full Historical DTD Output

The company's DTD history from 2 January 2024 through the same cutoff date as the parameters, for comparison and reflection. The parameter and output cutoff date is **15 September 2026**.

Output column	Description
`COMPANY_ID`	Company identifier
`REAL_DTD_DATE`	DTD calculation date, YYYYMMDD
`DTD_VALUE`	Historical production DTD

Part 1 — Build a Daily DTD Service

Build a locally runnable Python-based web service that integrates the following two modules—**data ingestion and preparation** and **DTD computation**—into a daily production workflow. Provide a simple interface to run the workflow and review its results. The service design and implementation approach are up to you.

Focus: Prioritize a clear, integrated workflow and reproducible results over custom algorithms or performance tuning. Standard numerical libraries are welcome.

A. Data Ingestion and Preparation

The supplied historical inputs end on **31 December 2025**. Update the dataset through the **most recent completed trading day** and state the cutoff date. Your web service should allow us to test daily updates for recent trading dates.

Input Updates

Support the following two types of updates through the web service, integrating new inputs with existing history without altering historical records:

1. **Daily inputs:** Automatically retrieve new market capitalization and risk-free rates (HKMA Exchange Fund Bills — 12 Month) for each trading day.
2. **Financial statement inputs:** Incorporate any necessary financial updates when extending the dataset; manual entry is acceptable. For testing recent daily updates, you may assume no new statements are released and carry forward existing financial inputs.

The workflow should support daily incremental updates. How you structure and demonstrate it is up to you.

Data Source & Automation Expectations

- Use **official or reputable public sources**.
- Prefer repeatable, programmatic access, such as APIs or scheduled downloads.
- Briefly document source URLs or APIs, transformations, assumptions and limitations.

Data Consistency and Quality

Keep the following in mind:

- **Consistency:** Keep historical and new inputs aligned in definitions, dates, currency, units and rate conventions.
- **Data quality:** Consider missing, invalid or duplicate observations, and briefly explain your treatment and any transformations.

B. DTD Computation

Scientific Background

DTD is a daily credit-risk signal derived from the Merton structural framework, which links a firm's equity value to its underlying asset value and volatility. In our daily workflow, newly ingested data is validated and combined with existing historical inputs. The system then uses the supplied asset volatility σ and other-liability weight δ to solve the implied asset value (AV) and compute the day's DTD. These results are stored and made available through the web service, allowing users to monitor changes in the firm's credit risk over time.

Daily Calculation Workflow

Using the prepared inputs from **A. Data Ingestion and Preparation**, use the supplied σ and δ to solve daily implied asset value (AV) and calculate DTD.

Use the following assumptions and considerations:

- $\tau = 1$ year; the supplied σ is annualized.
- Match σ and δ to the calculation date. For service tests beyond the supplied parameter coverage, you may carry forward the latest available values; identify results that use this assumption.
- Use only information available as of the processing date and account for data-quality issues identified during preparation.
- Briefly explain your numerical approach and convergence checks, and flag failed calculations.

DTD Methodology — Reference

The equations and notation below summarize the DTD framework for reference. See NUS-CRI Technical Report 2023, Section 3.2 for further model background. Parameter estimation is outside the scope of this task.

1. Default threshold

$$OL_t = TL_t - CL_t - LTB_t,$$
$$L_t = CL_t + 0.5\,LTB_t + \delta\,OL_t.$$

2. Implied asset value

$$E_t = V_t N(d_+) - L_t e^{-r_t \tau} N(d_-),$$
$$d_+ = \frac{\ln(V_t/L_t) + (r_t + \frac{1}{2}\sigma^2)\tau}{\sigma\sqrt{\tau}},$$
$$d_- = d_+ - \sigma\sqrt{\tau}.$$

3. Distance-to-Default

Under the drift simplification used for this task:

$$\mathrm{DTD}_t = \frac{\ln(V_t/L_t)}{\sigma\sqrt{\tau}}.$$

Notation

Symbol	Definition / Convention
E_t	Market capitalization
V_t	Implied asset value (AV)
r_t	Annual risk-free rate in decimal units
σ	Supplied annualized asset volatility, in decimal units
δ	Supplied other-liability weight
CL_t	Current liabilities
LTB_t	Long-term borrowings including non-current lease liabilities
TL_t	Total liabilities
OL_t	Other liabilities: $TL_t - CL_t - LTB_t$
L_t	Default threshold: $CL_t + 0.5\,LTB_t + \delta\,OL_t$
$N(\cdot)$	Standard normal cumulative distribution function
τ	Set to 1 for this task, representing a one-year horizon
DTD_t	Distance-to-Default at date t

Part 2 — Comparison and Event Study

Provide a short report covering the following two areas.

A. DTD Output Comparison

- Compare your computed DTD with CRI's supplied output over dates covered by both the supplied parameters and output.
- Discuss notable differences and their possible causes, including data sources, input preparation and numerical implementation.
- Treat CRI's output as a reference for analysis, not a target for fitting parameters. State assumptions and limitations.

B. Event Study – Reflection Questions

Event Identification

Based on public information, identify key events relevant to the firm's credit risk during the sample period, including any signs of credit distress. State event dates and cite your sources. You are not expected to be exhaustive.

Model Signal vs Reality

How does the historical DTD trajectory behave around these events?

1. Does it appear to anticipate, coincide with, or lag the observed changes in credit conditions?
2. Are there periods where the signal appears noisy, muted, or counter-intuitive?

We value clear reasoning, alternative interpretations and explicit limitations over a single definitive answer.

Submission

Expected time: approximately 8–12 hours.
Please prioritize **clarity of reasoning and system design** over completeness or sophistication.

Please submit a **GitHub repository link** containing the following, and ensure reviewers can access it:

1. **Code** demonstrating the integrated daily workflow:
    1. Clear structure and readable implementation.
    2. Dependencies, setup instructions and sufficient comments to make the service easy to review and rerun.
2. **Data** used and produced in the task:
    1. All supplied and newly collected data used in your solution.
    2. Prepared inputs and computed outputs, with sources and date coverage identified.
3. **Documentation: a short report** covering Part 1 and Part 2:
    1. Overall approach, workflow design, assumptions and limitations.
    2. Data sources, transformation decisions and key trade-offs.
    3. DTD output comparison and event study, with references.
    4. An **AI use statement**: tools used, how you worked with them, and how you reviewed and verified their outputs. We encourage AI use and want to understand your collaboration with AI; state if none was used.

We evaluate:

- How you structure and integrate a maintainable daily workflow.
- How you reason under uncertainty and communicate assumptions and trade-offs.
- How you connect financial understanding, data preparation and computation.

Clarity, sound judgment and a working end-to-end workflow matter more than technical sophistication.